CHAPTER 15

Ruby Gems

One of the tremendous benefits of being a Ruby developer is being able to take advantage of the entire ecosystem of other Ruby developers who have written and shared useful tools with the community. These tools are called *gems*, a term that means both the library itself, and the packaging structure that is used to manage them. Ruby comes standard with RubyGems, a command-line tool for installing and managing Ruby gems, and Bundler, a tool for creating manifests of gem versions so that developers all use the same set of dependencies.

Not only can you use existing gems, but also you can write your own. Even if you don't intend on sharing your gem, the basic structure of a Ruby gem is a good skeleton for your application. And if you do want to share your gem, you can do that via the central repository at http://www.rubygems.org.

Installing and Managing Gems

All the RubyGems tooling comes standard as a part of Ruby, including the command-line application `gem`, which you can use for many of your RubyGem-related needs. Ruby gems conform to a standardized format that provides metadata about the gems, and most importantly about any other gems that this gem might depend on. The `gem` command-line tool knows to look in the central repository for gems, but you can also point it to look at other sources. Using other sources allows you to maintain a private gem repository, for example as a place to keep internal tooling that you don't want made public.

In this section, we'll talk about managing gems directly from the command line, which is useful, but please keep in mind that on any application that is at all complex, much of the installation and gem management will be handled by Bundler, which we'll look at in Using Bundler to Manage Groups of Gems, on page 253.

Searching for Gems

Let's start by finding a gem to install. So far, the only gem we've discussed is RSpec, which is useful for our purposes. In order to download a gem, we need to know the exact name under which the gem is stored in the central gems repository. In this case, which is fairly typical, the internal name of the gem, `rspec` is just the lower case version of the colloquial name, RSpec.